EEE8085
M2M Technology IoT
Topic 2: The Internet

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Aims/Objectives

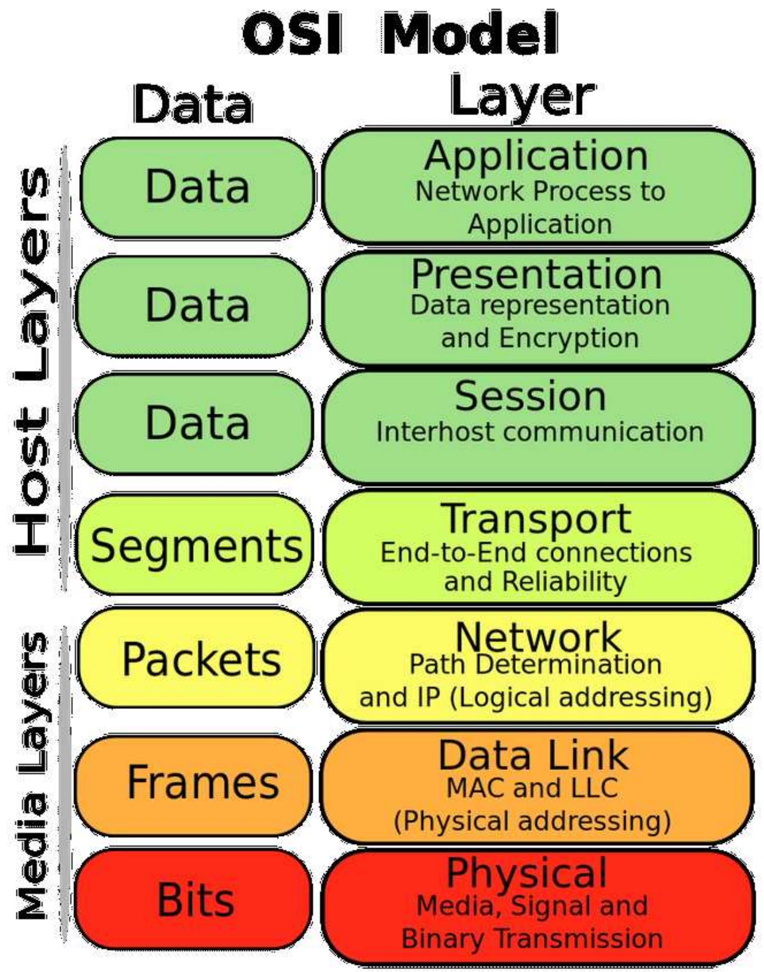
Aim

- The necessary background on the Internet, so we could later construct an M2M network on top of it.

Objectives

- Skip OSI level 1 “Physical”, as it is encapsulated into the off the shelf standard interfaces.
- OSI level 2 “Link” to see a couple of examples of MAC with the corresponding packet formats.
- OSI level 3 “Network” or IP protocol and LAN.
- OSI level 4 “Segments” or TCP and other protocols.

Open Systems Interconnection model

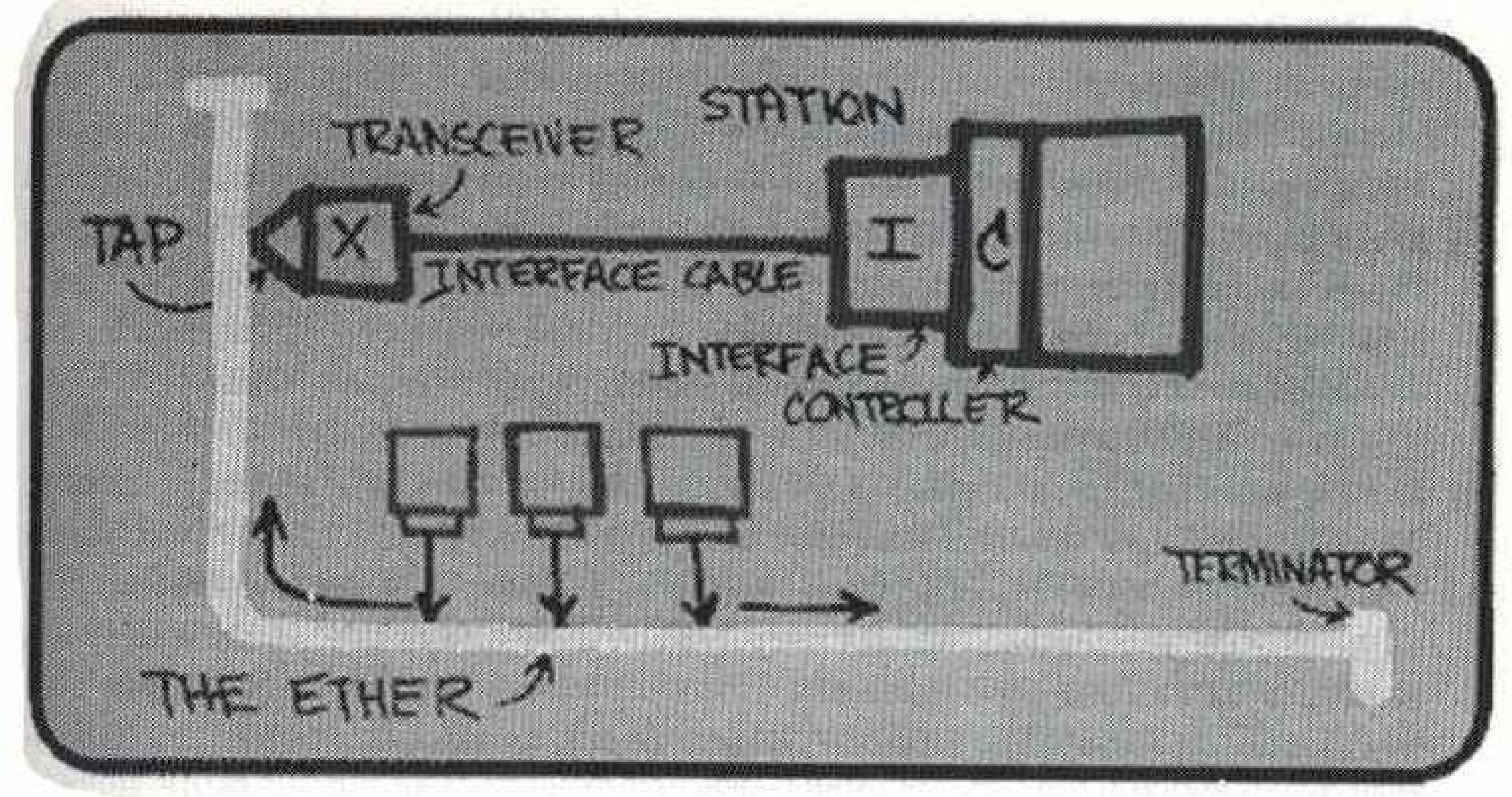


Link layer

The lowest logical level (Link or OSI 2)

- Ethernet standards IEEE 802, about 60 of them...
  - Ethernet II or DIX over thick coax (1982) – first usable
  - IEEE 802.3 10Base5 10Mbit/s (1.25MB/s) over thick coax (1983)
  - IEEE 802.3u 100BASE-TX over CAT5 (1995)
  - IEEE 802.3ab 1000BASE-T over CAT5e/6 (1999)
  - IEEE 802.3bz 2.5 and 5GBASE-T over CAT5/6 (2016)
  - may include PPPoE and PPP additional layers
- WiFi

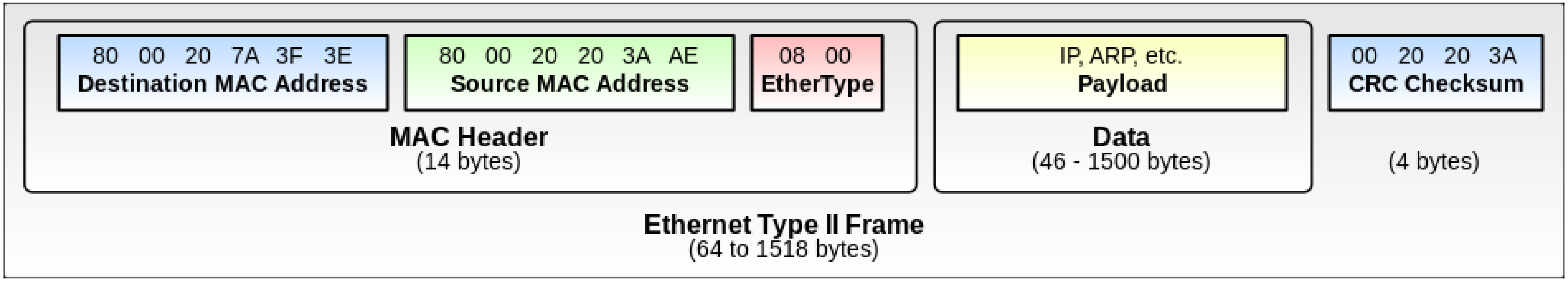
Ethernet invented



1970 – ALOHA protocol by Norman Abramson, U. of Hawaii
1973 – Ethernet prototype circuit board by Robert Metcalf, Xerox

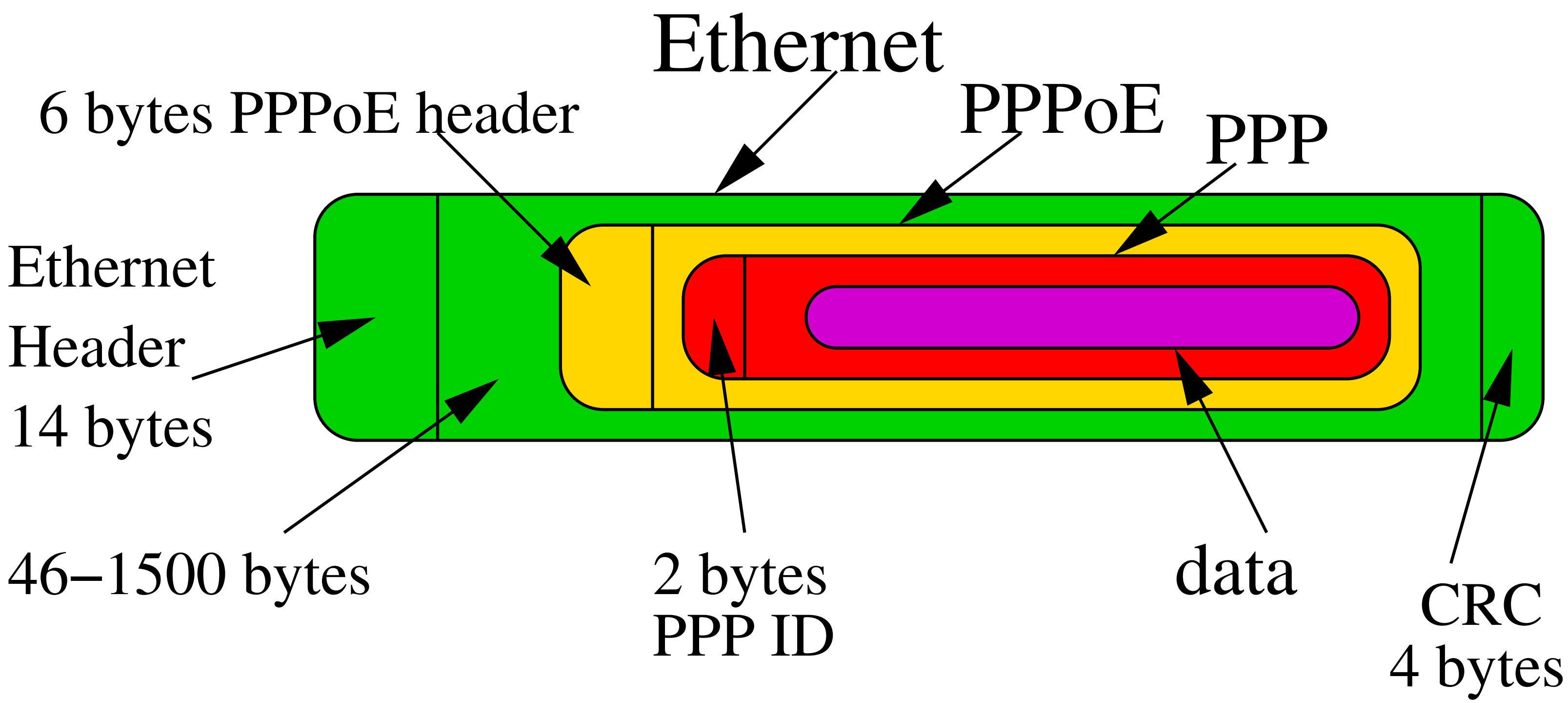
Ethernet frame

A packet of OSI 1 (physical), contains a frame of OSI 2 (link) Ethernet II frame



EtherType:
0x0800 – IPv4 packet in the payload
0x86DD – IPv6 packet in the payload
0x0806 – ARP packet in the payload
Ethernet II: Frame length is detected at the physical level by the hardware.
Ethernet 802.3: EtherType can be substituted with FrameLength
Details are here: http://www.ieee802.org/3/

Ethernet MTU



Due to use of PPPoE/PPP the value of MTU should be not greater than **1492** [RFC2516, 1999]
This is changed now to **1500** in [RFC4638, 2006] by adding additional requirements to hardware and a negotiation protocol.

WiFi Link-level packet

2	2	6	6	6	2	6		4
FC	DI	Addr.	Addr.	Addr.	SC	Addr.	Frame body	CRC

FC – Frame Control
DI – Duration/Connection Id
Addresses – depend on the context
SC – sequence control

- Frame body – data unit or control information, size 0-...:
  - WEP : 2304 + 34 + 8 = 2346 bytes
  - WPA (TKIP) : 2304 + 34 + 20 = 2358 bytes
  - WPA2 (CCMP) : 2304 + 34 + 16 = 2354 bytes

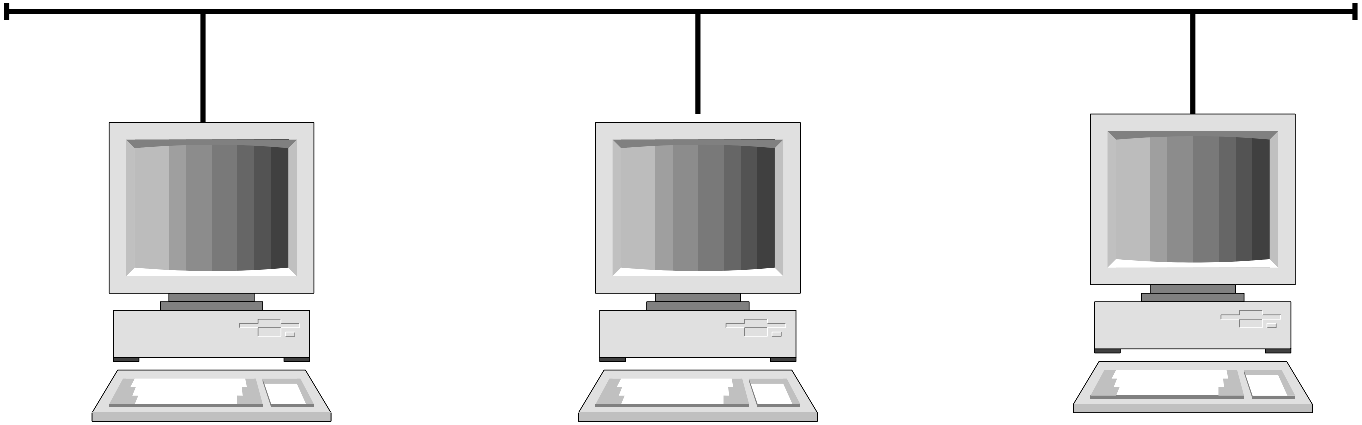
Details are here: http://www.ieee802.org/11/

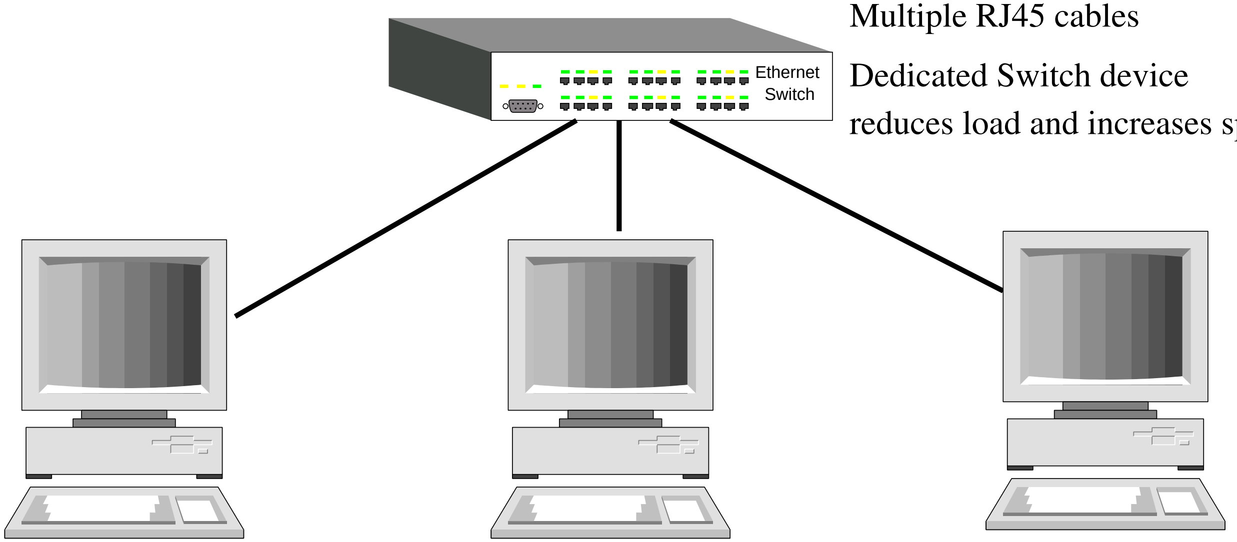
Common features of link layer

- Source and destination MAC addresses.
- MAC address format: 6 bytes, $2^{48} = 281474976710656$ combinations.
- MAC address of the device is usually assigned at the factory, but can be changed.
- Payload – data
  - directly included,
  - or encapsulated into PPPoE/PPP. In this case Ethernet+PPPoE/PPP stack is considered a Link layer PDU.
- Carry a datagram (packet) between two IP hosts
- Various devices on the link allow transition between different instances of media

Examples of Link layer (1)

Single coax cable with multiple taps, old EEE802.3

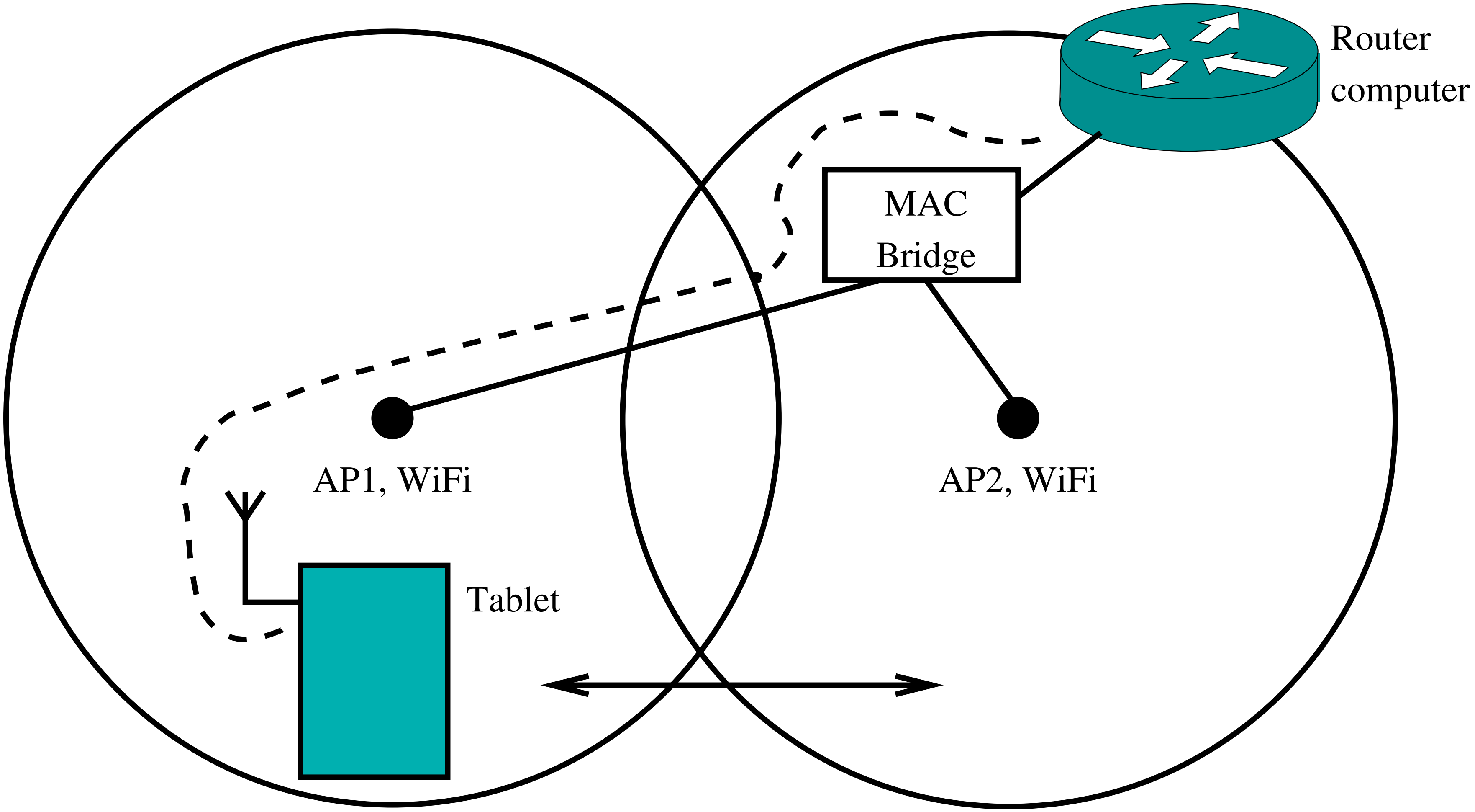




Multiple RJ45 cables

Dedicated Switch device
reduces load and increases speed

Examples of Link layer (2)



Capture Link-layer packets

Simple! Type this, then read the manual and adjust the options

- tcpdump -e -n -i eth0 -XX

You may try to substitute eth0 with any other interface discovered by

- ifconfig

To save the dump in a file type

- tcpdump -e -n -i eth0 -XX > dump-eth0-1.txt

You may want to filter out the lines containing your MAC address

- tcpdump -e -n -i eth0 -XX | grep ???? (several digits of eth0 MAC in hex)

Network layer

IP – Internet Protocol

- it sends packets or datagrams from one point to another
- uses the Link layer protocol
- IP address is SOFT as opposed to the hardware MAC address

ARP – Address Resolution Protocol

- converts IP–>MAC address
- broadcasts ARP requests on the network
- receives replies containing MAC and IP addresses

ICMP - Internet Control Message Protocol

IP address and netmask

IPv4 – 4 bytes, e.g. 192.168.0.1

netmask – also 4 bytes, e.g. 255.255.0.0 or /16 (the number of 1-s from the left)

- IP address identifies a computer on the web
- Together with a netmask an IP address identifies a range of addresses
  - “192.168.0.0/16” or “192.168.0.0 netmask 255.255.0.0”
- Such a range defines a LAN
- Such a range can be used in a routing table to specify a gateway per range

Classes of IP addresses

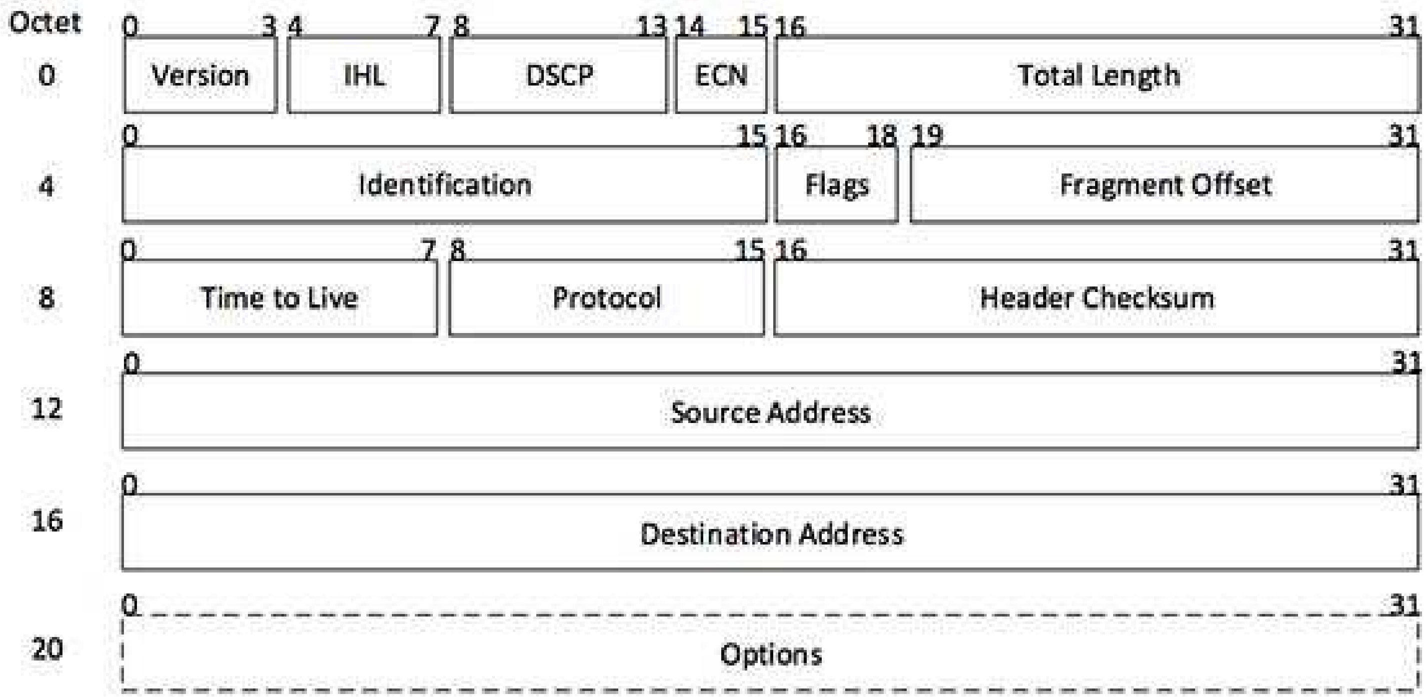
Network Class	First Octet	Network Mask
A	1. to 127.	255.0.0.0
B	128. to 191.	255.255.0.0
C	192. to 233.	255.255.255.0
D	224. to 239.	None

- IPv4
  - 4 octets – 4G addresses
  - Is it enough?
- IPv6
  - two more octets
  - painful transition

Private ranges

- Class A: 10.x.x.x
- Class B: 172.16.0.0 – 172.31.255.255
- Class C: 192.168.x.x

IP packet structure



[Image: IP Header]

No MAC address – it sits inside a Link-layer packet.
Fragmentation may happen if the Link MTU is small.
Time to Live is used in “traceroute”
https://www.tutorialspoint.com/ipv4/ipv4_packet_structure.htm

ARP

If a computer does not know the MAC address of the target (another computer on LAN of Gateway) it sends an ARP request, which is a link-layer packet with destination FF:FF:FF:FF:FF:FF. This packet carries an ARP packet.

- Request a MAC address for an IP address
- ARP probe – checking in an IP address is already in use
- ARP announcement – tell everyone your IP address
- other ARP functions...
- ARP cache (can be filled from a file for security)
  - IP addr., MAC addr., HW type, interface, flags

Transport protocol

- IP is the bedrock of TCP/IP.
- Every piece of data is sent as an IP packet.
- IP transmits packets across and between networks
  - hence the name: INTERNET
- TCP – Transmission Control Protocol
  - splits the message into pieces (datagrams)
  - the datagrams are delivered by the IP
  - reassembles the message from datagrams
  - requests resending lost datagrams

Datagram structure (formed by TCP)

<table>
<tr><td colspan="3">Source port</td><td>Destination port</td></tr>
<tr><td colspan="4">Sequence number</td></tr>
<tr><td colspan="4">Acknowledgement number</td></tr>
<tr><td>Data offset</td><td>Reser−ved</td><td>Flags</td><td>Window</td></tr>
<tr><td colspan="3">Checksum</td><td>Urgent pointer</td></tr>
<tr><td colspan="4">your data ... next 500 octets</td></tr>
</table>

Not looking into it in-depth, because our chosen software operates it automatically.
See the tutorial link on Blackboard.

How it works (1)

- Addresses
  - Ethernet address is unique per card
    - H/W or MAC address (Media Access Control)
  - IP address must be either manually assigned or by DHCP
  - Several network interface cards
  - An interface can have several IP addresses (multihoming)
- Subnet masks
  - splits the IP address into network and host parts
- Gateways
  - direct access to networks beyond the given one

How it works (2)

- Names
  - people work better with words than with numbers
  - a name corresponds to an address
  - several names can point at the same address (alias)
- Setting the name-IP address correspondence
  - file /etc/hosts
    - simple, but every host needs to be configured
  - name server
    - a hierarchy of name servers
    - automated database update

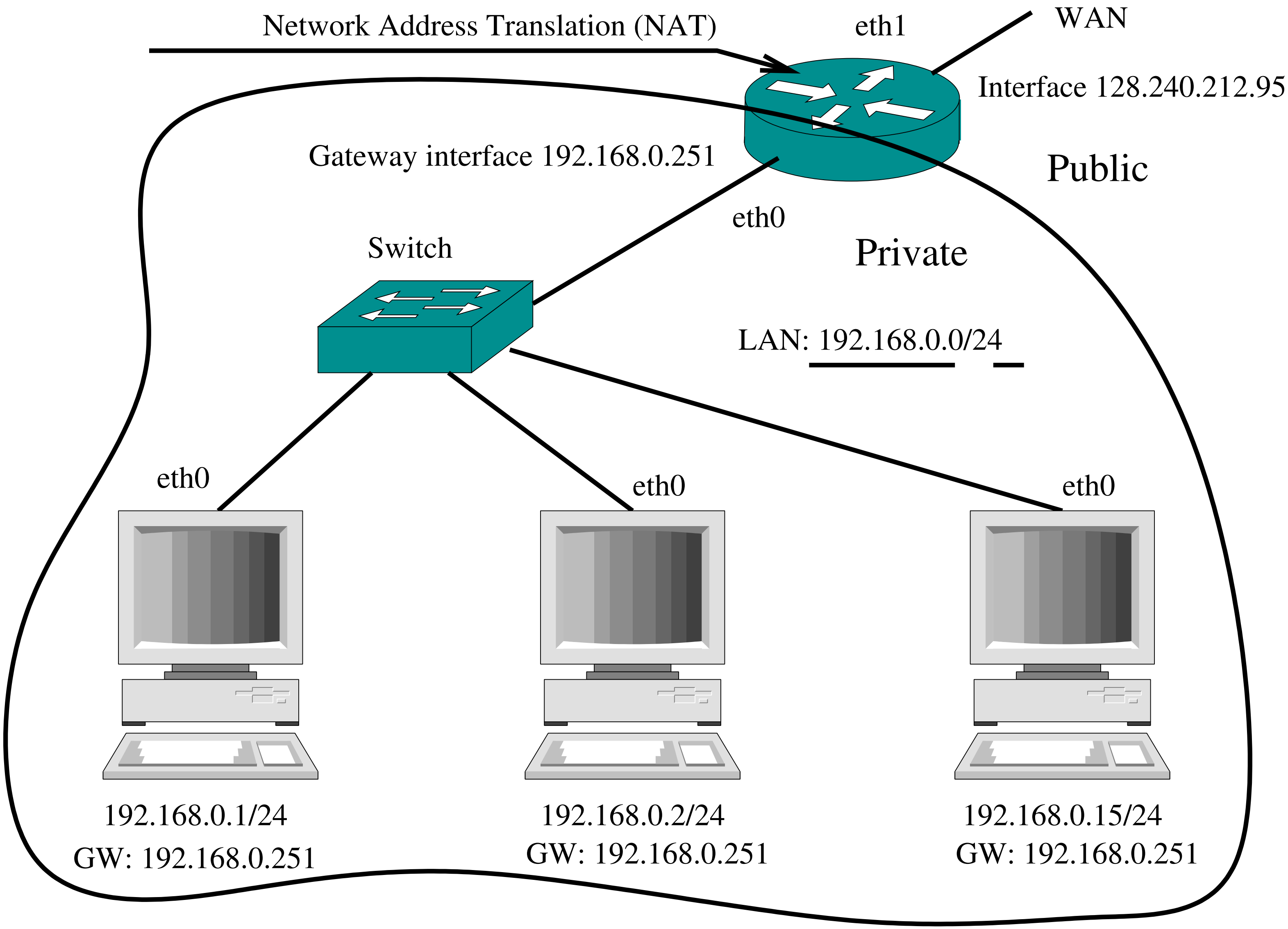
How it works (3)

- LAN – every host sees all packets send by other hosts (Ethernet)
- Router
  - compares the destination address with the table of addresses in its memory
  - if the match is found, it forwards the packet to the corresponding address
  - the destination can be
    - address at another network
    - address of the next hop router
  - if the match is not found
    - send it to the Gateway router (if there is one)

How it works (4)

- Gateway – advanced router
  - receives the traffic whose destination is unknown to the LAN
  - LAN computers talk to each other directly
- static routing
  - the routing table is read from a file
  - adequate for simple networks
- dynamic routing
  - too complex for this course
  - based upon broadcasting ICMP (Internet Control Message Protocol) router solicitation messages
  - routes are updated periodically in response to traffic conditions

How it works – simple example (1)



How it works – simple example (2)

- The sending application program prepares its data and calls on its local internet module to send that data as a datagram and passes the destination address and other parameters as arguments of the call.
- The internet module prepares the PDUs nested inside each other
  - UDP or TCP packet containing to ports on the source and destination ports,
  - IP packet containing the source and destination IP addresses, plus the routing information
    - to LAN, to Gateway, to a LAN specified in the routing table.
  - MAC packet containing the source and destination MAC addresses (ARP cache is used)
- It sends this packet through the local interface chosen

How it works – simple example (3)

- If the destination MAC is the IP destination, then just unpack the multiple PDUs and get the datagram.
- If the destination MAC is gateway/router, then unpack the MAC layer, and try to send the IP packet further by the same procedure as above.

Linux box basic configuration

Enable IPv4 routing and disable bypassing LAN hack:

```
/etc/sysctl.conf:net.ipv4.ip_forward=1
/etc/sysctl.conf:net.ipv4.conf.all.accept_redirects = 0
/etc/sysctl.conf:net.ipv4.conf.all.send_redirects = 0
```

Configure an interface /etc/network/interfaces:

```
auto eth0
allow-hotplug eth0
iface eth0 inet static
address 192.168.0.1
netmask 255.255.255.0
gateway 192.168.0.251
```

Linux box NAT configuration

#Masquerade to wlan0

iptables -t nat -A POSTROUTING -s 192.168.0.0/24 -o wlan0 -j MASQUERADE

#List NAT configuration

iptables -t nat -n -L

#Delete the masquerade

iptables -D POSTROUTING -t nat -s 192.168.0.0/24 -o wlan0 -j MASQUERADE

Creating custom gateways

- Normally, all non-LAN destination packets are sent to a default gateway
- Custom gateway – send a range of IP addresses to a particular router on your LAN
  route add -net 192.168.2.0 netmask 255.255.255.0 gw 192.168.0.2
- Chaining such gateways is not easy. Don't forget to configure the return path!

Experiments on the live net

- ping
  - sends a special packet to the IP destination and receives a reply
- traceroute
  - Probes the route to the IP address
  - sends a series of UDP packets with different TTL values
  - the TTL is decremented in each hop
  - when TTL==0 an error message is returned carrying the address of the last router
- tcpdump – inspects the packets arriving to an interface